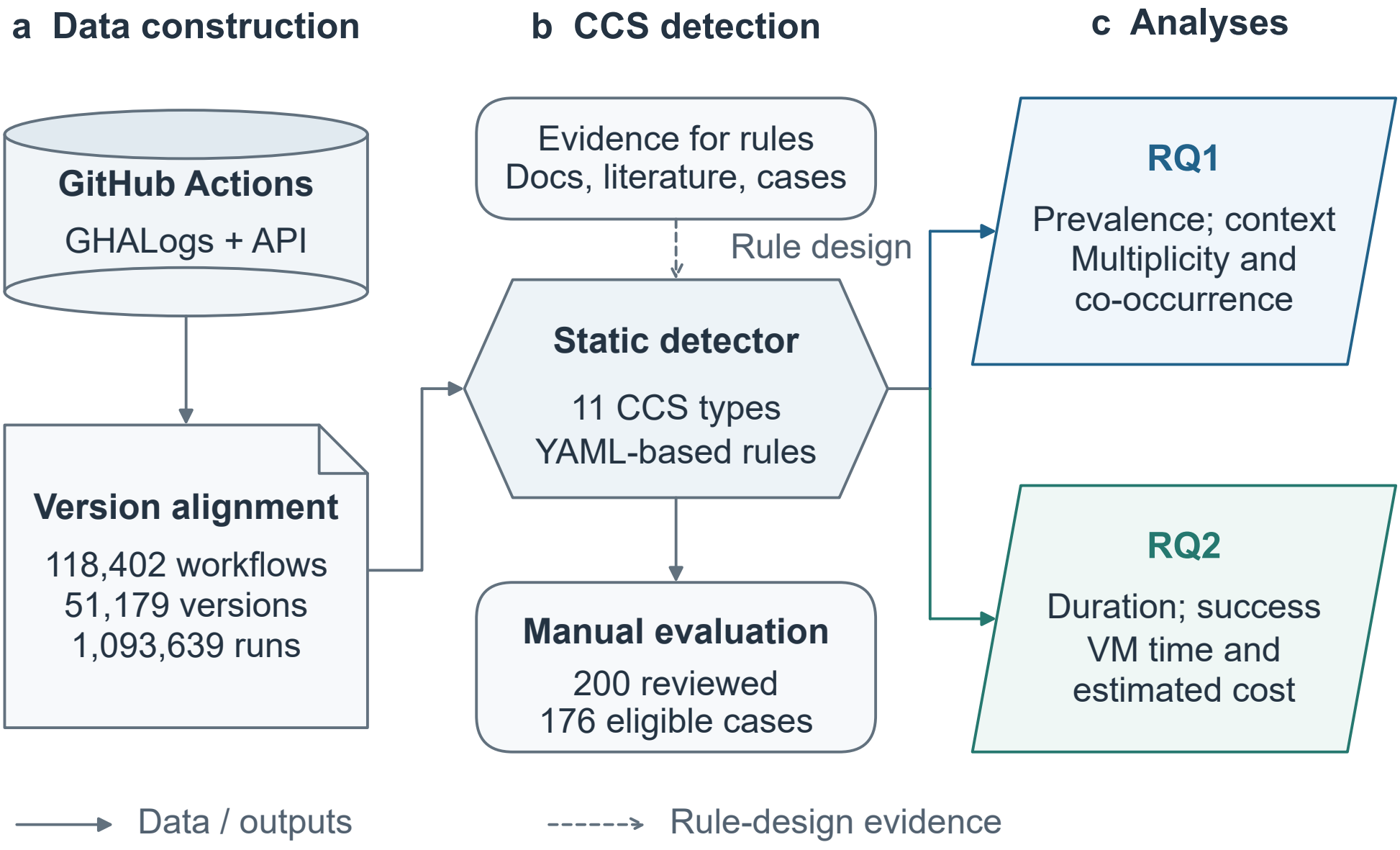
a Data construction
b CCS detection
c Analyses
GitHub Actions
GHALogs + API
Version alignment
118,402 workflows
51,179 versions
1,093,639 runs
Evidence for rules
Docs, literature, cases
Rule design
Static detector
11 CCS types
YAML-based rules
Manual evaluation
200 reviewed
176 eligible cases
RQ1
Prevalence; context
Multiplicity and
co-occurrence
RQ2
Duration; success
VM time and
estimated cost
Data / outputs
Rule-design evidence